

Applied Ontology Introduction + Upper-Level Distinctions

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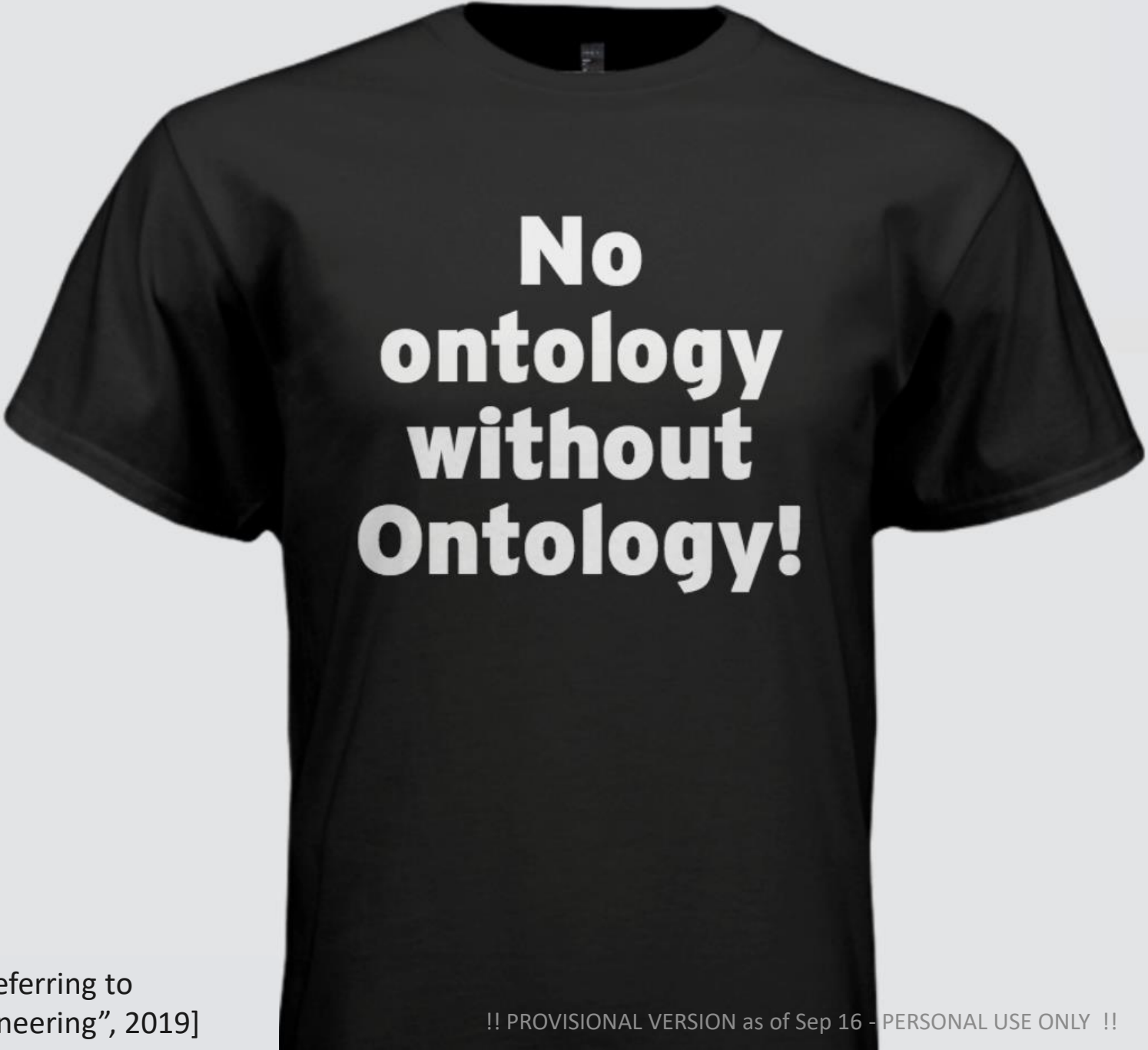
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A black t-shirt is displayed against a light blue background. The t-shirt has a crew neck and short sleeves. Centered on the chest is the text "No ontology without Ontology!" in a bold, white, sans-serif font. The word "No" is on the first line, "ontology" is on the second line, "without" is on the third line, and "Ontology!" is on the fourth line.

**No
ontology
without
Ontology!**

Agenda

- **Applied Ontology (AO)**
 - What is Applied Ontology?
 - AO History
 - What Are Ontologies for?
 - Interdisciplinarity of AO
- **Ontology & -gies**
 - What is (an) Ontology?
 - Types of Ontologies
 - What is Formal Ontological Analysis?
- **Upper-Level Distinctions**
 - Class, Individual
 - Class, Relation, Role
 - Instantiation, Subclass, Taxonomy
 - Space, Time, Concrete, Abstract
 - Continuant | Endurant, Occurrent | Perdurant, “Snapshot”
 - Quality, Object, Inherence
 - MaterialObject, AmountOfMatter, Feature
 - ...

Disclaimers

my perspectives only – coming from knowledge representation/computer science

please ask, criticize and comment (any time!)

apologies for any dense slide

What is Applied Ontology (AO)?

something that can be handy for real-world affairs:

- One or two **events**?
 - court case due to Sep 11, 2001 attack at twin towers
 - decision worth 3.5 billion USD, depending on understanding “occurrence”
- How many **buildings** does Germany have?
 - nobody *can* know, lacking a clear, broadly accepted definition of building
 - relevance: statistics determine allocation of money
- note already: **meanings** are involved
 - not at least **of terms/words/lexical units**



What is Applied Ontology?

- 1) an **interdisciplinary field** dealing with **ontologies, ontological analysis & conceptual modeling** and **applications thereof**
 - broad overlap with information and computer sciences
 - initially called **Formal Ontology in Information Systems (FOIS)**
 - differs (with overlap) from sub areas of philosophy of Applied Ontology and Formal Ontology
- 2) an **activity** of applying ontologies and methods of those field(s)
- 3) the major **journal** of the field (1)



What is Applied Ontology?

an interdisciplinary field dealing with ontologies, **ontological analysis** & **conceptual modeling** and applications thereof

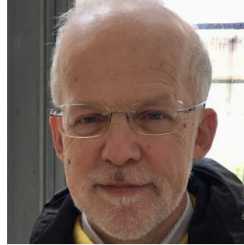
The former includes any attempt to **investigate the nature and structure of a domain of interest** using **rigorous philosophical or logical tools**; **the latter** concerns the **cognitive and linguistic** structures we use to **model the world**, as well as the various analysis tools and methodologies we adopt for **producing useful computational models**, such as information systems schemes or knowledge structures.

[excerpt of the description of the journal Applied Ontology]

AO History

3 central figures & their groups

- **Nicola Guarino**
+ Univ. of Padua
+ LOA (Laboratory of AO, Italy)
- **Barry Smith**
+ IFOMIS (Germany)
+ Univ. of Buffalo (USA)
- **Giancarlo Guizzardi**
+ NEMO, Fed. Univ. of
Espírito Santo (UFES, Brazil)
+ Univ. of Twente (Netherlands)

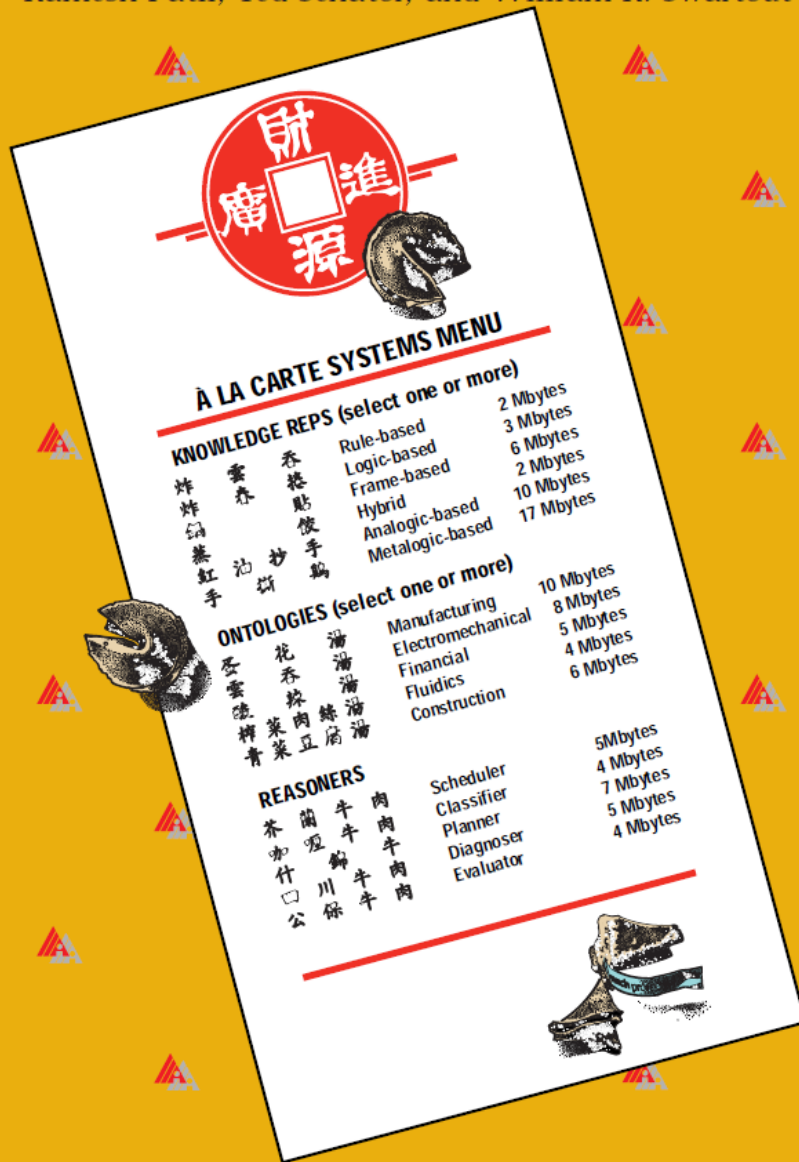


1st bits of history

- 1993 – Padua Workshop
- 1998 – 1st **FOIS** (Trento)
- 1998/2000 – Gene Ontology
- 2001 – Semantic Web vision/article
- 2002-2004 – WonderWeb project
- 2003 – LOA (Laboratory of AO)
- 2005 – **journal AO** launched
- 2009 – **IAOA** founded
- 2009 – 1st ICBO (Buffalo)
- 2015 – 1st **JOWO** @ IJCAI (Buenos Aires)
- 2025 – 15th FOIS (Catania)

Enabling Technology for Knowledge Sharing

Robert Neches, Richard Fikes, Tim Finin, Thomas Gruber,
Ramesh Patil, Ted Senator, and William R. Swartout



“Enabling Technology for Knowledge Sharing”,
Neches et al., [1991](#)

Another look at data

1967

by GEORGE H. MEALY
Computer Consultant
Scituate, Massachusetts

INTRODUCTION

particular ontology, we can avoid a quarrel by adopt-

*“data are fragments of a theory of the real world, and data processing juggles representations of these fragments of theory...**The issue is ontology, or the question of what exists.**”*

them in a somewhat new form may prove to be at least suggestive.

To begin on a philosophical plane, let us note that we usually behave as if there were three realms of interest in data processing: the real world itself, ideas about it existing in the minds of men, and symbols on paper or some other storage medium. The lat-

Toward a theory of data

Relations

To fix our ideas, consider the following example of genealogical data, taken from Reference 2:

What Are Ontologies For?

semantic interoperability was **the original** purpose and is still **a most important** one of ontologies

- whether framed as
 - **data** integration
 - reuse of **information**
 - **knowledge** sharing
 - facilitating shared understanding among humans (and/or machines)
 - **ontology** integration
- with **various purposes** (until **today** in general, e.g.
 - semantic search, retrieval
 - data annotation
 - backbones for knowledge graphs & linked data

Interdisciplinarity

- let's return to: What is Applied Ontology?

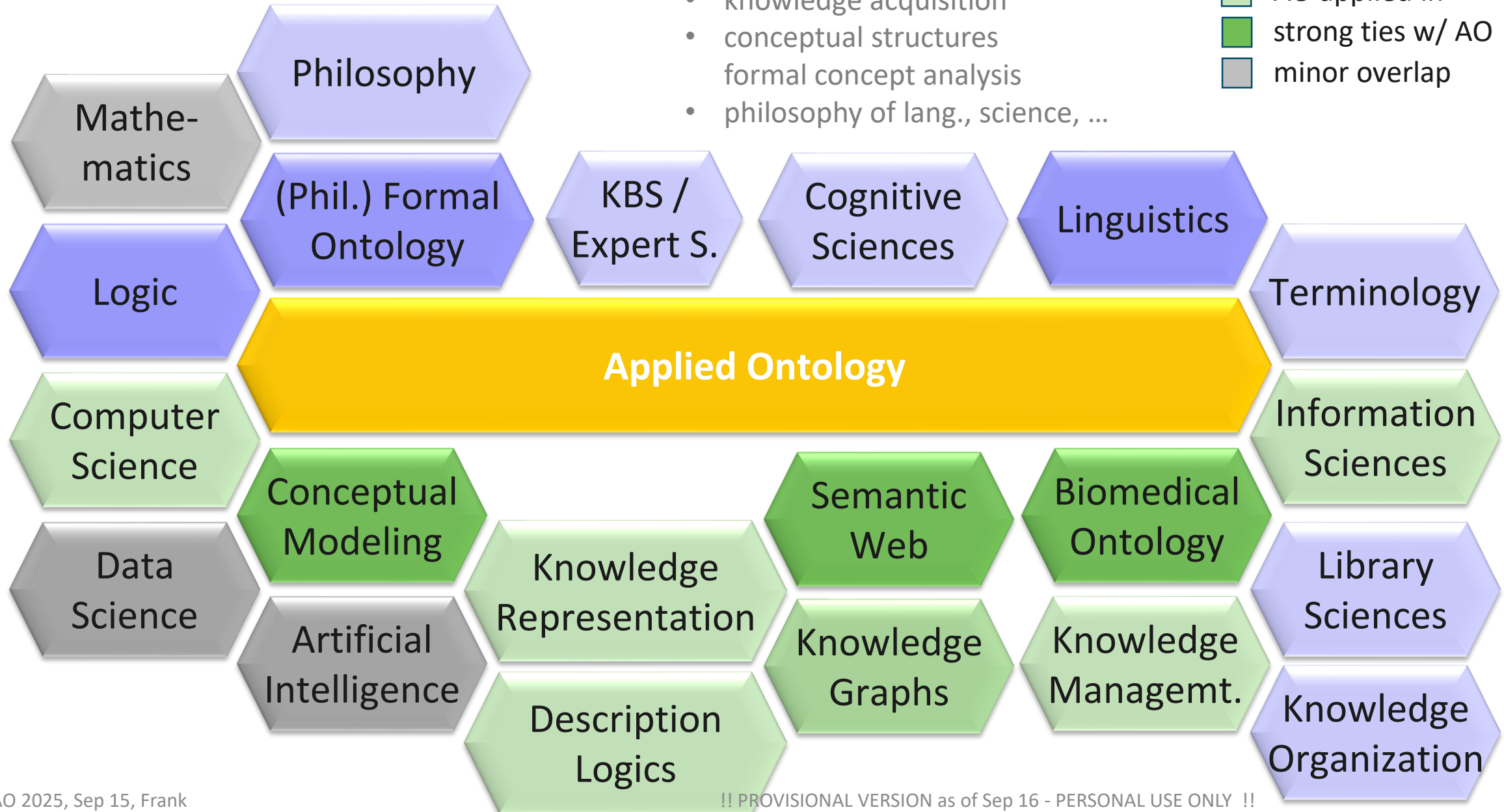
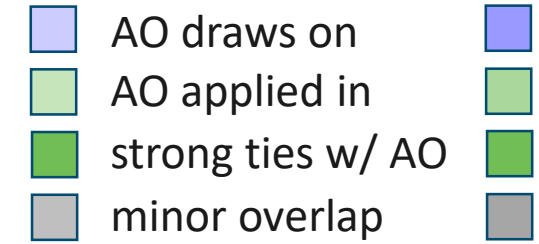
an **interdisciplinary** **field** dealing with **ontologies**, **ontological analysis & conceptual modeling** and **applications thereof**

Interdisciplinarity

@ISA025

for lack of space not integrated:

- knowledge engineering
- knowledge acquisition
- conceptual structures
- formal concept analysis
- philosophy of lang., science, ...



Interdisciplinarity Means What?

a few more precisely

- philosophical formal ontology
 - adoption of formal ontological tools
- linguistics
 - semantics as study of meaning
 - lexical semantics – word meaning
- logic, DL, conceptual modeling
 - How to represent ontologies?
 - What's the interplay between formal semantics and ont. content?

beware of **varying terminology**

- class
 - OWL: set of individuals, subject to axioms
 - OOP: template for instances, w/ attributes and methods
 - UML: set of objects sharing attributes, operations + constraints
 - math./logic: a set-theoretic notion for „too large sets“ (e.g., the class of all sets)
- category
 - philosophy: a most general/highest kind
 - cognitive science: mental representation/grouping of entities
 - AO: an alternative for class
- {class, type, kind, concept, category, term, universal}
 - „synonymously“ usable in loose exchange
 - for each exist sub areas with a technical meaning

Ontology & -ies

- characterization & representation
- formal ontological analysis

What is an Ontology?

think of this as being **representation-agnostic**

- **typical ontology constituents**
 - **classes** / categories / types / concepts
 - **relations** / associations / object properties
 - attributes / datatype properties
 - individuals ?

below e.g.

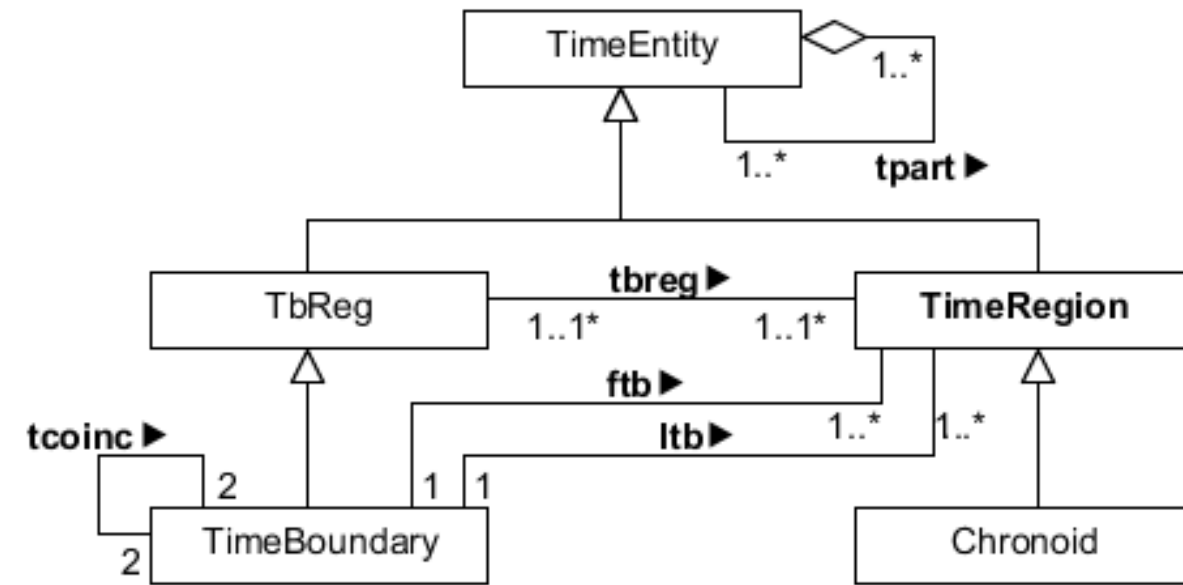
→ TimeEntity, TimeRegion

→ tpart, tbreg, tcoinc

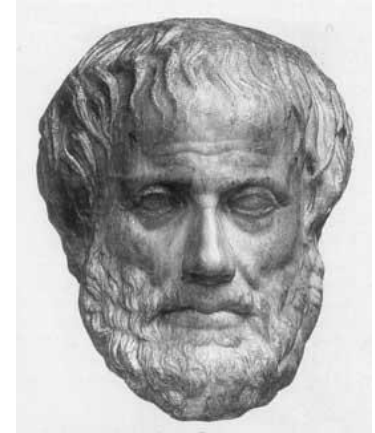
- **interconnections** of constituents
via **axioms** / rules / constraints

e.g.

```
:TimeRegion rdfs:subClassOf :TimeEntity
```



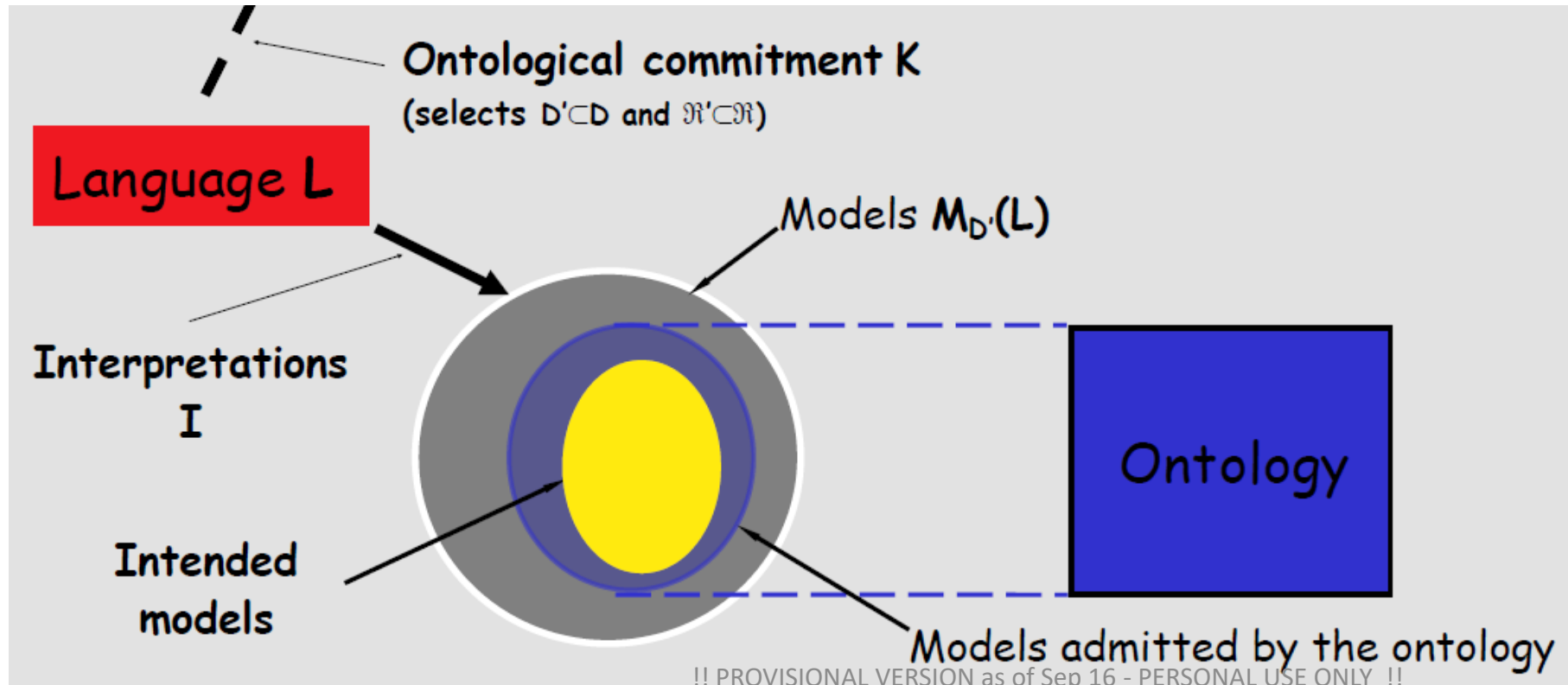
Prelude: Ontology vs ontology



- Ontology (with capital O)
 - the study of being / the science of what is
 - What exists?
 - What does it mean to exist?
 - Which are the most general kinds of entities?
 - a sub field of philosophy, relates to metaphysics, epistemology, etc.
- ontology (with lower-case o) in applied ontology
 - questions for ontology development
 - given D (a domain/database/system/task/...): What exists in D?
 - given an entity E from D: What is (an) E?

What is an ontology? → Definitions

- simplified version based on [Guarino, 2009]
a **logical theory** such that its formal models approximate the **intended meaning of a vocabulary** as good as possible



What is an ontology? → Definitions

- simplified version based on [Guarino, 2009]
a **logical theory** such that its formal models approximate the **intended meaning of a vocabulary** as good as possible
- [Studer, 1998]
a **formal, explicit specification** of a **shared conceptualization**
- [OWL 2 Specification, 2012]
basically a **set of axioms** (seen as true in the respective domain)
- [ISO 21838-1, 2021, 3.14 – ontology]
collection of terms* (3.7), relational expressions (3.6) and associated **natural-language definitions** (3.8) together with one or more **formal theories** (3.11) designed to capture the **intended interpretations** of these definitions
- [my view] (deliberately very wide)
a **theory** about **terms*** (and their interconnections) (in a context/domain)

* for me, terms include relational expressions and intended meaning

Types of Ontologies

- assume: „a **theory** about **terms** (and their interconnections)”
- **theory** → by degree of formalization
- **terms** → by level of generality
 - by domain
 - e.g. **biomedical, geo-, legal, business ont.**
- as a whole → by purpose
- historic remark: WordNet, a lexical database for English, was also referred to as linguistic or as lexical “ontology”

Types of ‘Ontologies’: By Degree of Formalization

- informal
 - term list, (controlled) vocabulary, catalog, glossary
 - natural language texts
- term hierarchies
 - taxonomy, nomenclature, classification
 - e.g. **ICD**, **UNSPSC**, ...
- semi-formal
 - thesaurus, semantic net, UML- or ER-diagrams
 - e.g. **FMA**, **MeSH**, **WordNet**, ...
- formal
 - logical theory, e.g. in FOL, OWL, ...
 - e.g. **SNOMED-CT**, (**GALEN**)

‘fully proper’ ontologies
in the sense of AO, if based
on formal ontological tools

informal

light-weight

heavy-weight

formal

Types of Ontologies: By Level of Generality

- top-/upper-level ontology (TLO) e.g. BFO, DOLCE, GFO, UFO, YAMATO
 - classes for most domains
 - e.g.: *object – process – quality*
- core ontology (CO) / mid-level ontology (MLO) e.g. CCO
 - classes for a larger field/discipline
 - e.g.: *human – disease – severity*
- domain ontology e.g. most OBO Foundry ontologies
 - classes for a specific domain
 - e.g.: *leucocyte – chronic lymphoid leukemia*

Types of Ontology: By Purpose

reference ontology

- **purposes:**
 - (re)use as domain representation
 - enable interoperability across systems
- **meant to be:**
 - community-agreed
 - reliable, accurate, complete
 - canonical, irrespective of use cases and implementation detail
- **e.g. GO, SNOMED-CT, FIBO**

application ontology

- **purpose:**
 - enable/support a specific task or application
- **meant to be:**
 - narrow, specific scope
 - directly usable (for its use case), possibly pragmatically engineered
 - possibly derived from a reference o.
- **e.g. for ont.-based data access**

Formal Ontological Analysis



Nicola Guarino [ISAO 2023]

- defines **ontological analysis** as:
„a systematic way to understand and make explicit the world assumptions behind a certain description“ e.g.:
(1) This rose is red. (2) John is married with Mary. (3) John is a student.
- strongly argues for
 - **precision:** capturing intended meaning as exactly as possible
 - **accuracy:** choosing domain and modelling primitives appropriately
 - **well-foundedness:** using **general** categories and relations describing the nature and structure of a domain of discourse

Interoperability is not compatible with **underspecification!**

Formalized vs Formal Ontology

- **a formalized ontology**

is an ontology expressed in a formal (representation) language

- often a logical/logic-based language e.g. first-order logic (FOL) / OWL
- ,formal‘ referring to a formal semantics of the language

- **Formal Ontology**

- ,formal‘ here refers to Edmund Husserl and his „Logical Investigations“
- two aspects/meanings:
 - rigorous (e.g via formalization)
 - **general and neutral** wrt topic/entity considered, determined a priori
- concerns ,interconnections of things‘ like logic ,interconnections of truths‘
[Husserl, 1900-1901/1970, p.225, cited in Varzi, 2019]

Formal Tools for Ontological Analysis

per each item: **the theory of**

- parts (mereology)
- unity and plurality
- essence and identity →
- dependence
- composition and constitution
- properties and qualities
- time, persistence and change

→ cf. „Mereology“ on Tue

- ‚interconnections of things‘ ex.:
if $a = b$ and $b = c$, then $a = c$
(irrespective what a-c are)
- like logic: implication-transitivity

→ cf. „Time, ... Events“ on Tue

Ontology & -ies: Takeaways

PROVISIONAL VERSION DURING ISAO 2025, FOR PERSONAL USE OF REGISTERED PARTICIPANTS
(as of Sep 16: additions, esp. references of contributions to come)

- ontologies in Applied Ontology
 - an **ontology**: „a **logical theory** such that its formal models approximate the **intended meaning of a vocabulary** as good as possible” [Guarino]
 - major **constituents**: classes, relations and axioms about their interplay
 - key **purpose**: **semantic interoperability** → **no underspecification**
 - **distinctions** between
 - an ontology vs Ontology
 - formalized vs formal ontology (à la Husserl/Guarino)
- **types** of ontologies (in AO and neighbouring fields), including **top-level/foundational ontologies** and reference ontologies